BRANDSHIELD AI — MASTER SUBMISSION OUTLINE

HacXLerate 2026 | Challenge 03: Digital Risk Protection Platform (Presented by Kampus.vc)

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Challenge Track: Challenge 03 — Brand Impersonation & Rogue App Detection	**Submission Date:** October 13, 2026

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1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Fast-growing startups, fintech platforms, and educational institutions (such as ByteXL, Zomato, Zerodha, and universities) face a relentless wave of digital impersonation. Fraudsters deploy spoofed social handles, clone logos, and publish rogue APKs to defraud users via fake job registrations, UPI payment scams, and phishing links.

- **Fake Social Accounts:** Scammers create deceptive handles (e.g., *@bytexl_support_official*, *@zomato_refunds_care*) on Instagram, X, and Telegram.
- **Typosquatting & Homoglyphs:** Attackers substitute lookalike characters (*paytm* → *paytrn*, *0* for *o*, *1* for *l*) to bypass naive keyword filters.
- **72+ Hour Manual Delay:** Existing brand protection relies on slow manual reporting after victims have already lost money.

Our Solution — BrandShield AI: An autonomous, multi-modal Digital Risk Protection (DRP) platform that scans suspect handles and apps in real time, computes an objective 0–100 Unified Threat Index using **Meta FAISS vector similarity**, **NLP scam intent classification**, and **algorithmic string heuristics**, and generates **1-Click Automated Legal Takedown Notices**.

2. THE 3-PILLAR MULTI-MODAL AI DETECTION ARCHITECTURE

Pillar	Core Technology & Format	Detection Mechanism & Fraud Signatures Caught	Weight
Pillar 1: Vision Logo & Avatar Matching	Meta **FAISS** (*IndexFlatIP*) + **OpenAI CLIP / MobileNet** Exported to **.onnx** & **.index**	Converts official brand logos into 512-dim normalized dense embeddings. Performs cosine similarity search in **<15ms**. Catches cropped logos, altered color tints, watermarks, and fake verification badges.	**40%**
Pillar 2: NLP Scam Bio & Intent Classifier	**DistilBERT / RoBERTa** (**.onnx** / SafeTensors) + Fraud Pattern Rules	Detects financial solicitation (UPI handles like *@okhdfcbank*, *'Send Rs. 500 registration fee'*), urgency triggers (*'Immediate refund'*, *'DM for job'*), and shortened phishing links (bit.ly, raw IPs).	**35%**
Pillar 3: Heuristics Typosquatting & Metadata	**Levenshtein Distance** + Homoglyph Matrix + Affix Scanner	Computes character edit distance between official and suspect handles, detects homoglyph swaps (*rn→m*, *1→l*), flags scam suffixes (*_support*, *_official_india*), and checks follower/age anomalies.	**25%**

3. MATHEMATICAL THREAT SCORING ENGINE (0 - 100 SCALE)

Threat Score = (0.40 × S_visual) + (0.35 × S_bio_nlp) + (0.25 × S_handle_heuristic)

Score Range	Severity Level	UI Status Badge	Automated Platform Response
0 – 39%	**LOW / SAFE**	Emerald Green (#10B981)	Verified partner, legitimate fan page, or unrelated entity. Logged to audit history.
40 – 69%	**MEDIUM RISK**	Cyber Yellow (#EAB308)	Ambiguous handle or partial similarity; queued for human analyst review.
70 – 89%	**HIGH RISK**	Neon Amber (#F97316)	Probable brand impersonation; automated alert dispatched to security team.
90 – 100%	**CRITICAL THREAT**	Crimson Red (#EF4444)	**Confirmed Impersonation Attack:** Unlocks 1-Click Legal Takedown & Forensic PDF Report.

4. THE KILLER FEATURE: 1-CLICK AUTOMATED LEGAL TAKEDOWN

Enterprise security teams do not just want threat alerts—they need immediate resolution. When a CRITICAL threat (≥90%) is detected, BrandShield AI compiles an instant **Cease & Desist / DMCA Infringement Package** containing:

1. **Forensic Evidence Log:** UTC timestamps, suspect URL, handle ID, and side-by-side logo comparison with FAISS confidence score.
2. **Statutory Legal Citations:** Trademark infringement and impersonation clauses under the Information Technology Act & Platform ToS.
3. **Direct Abuse-Desk Dispatch:** Pre-addressed enforcement templates ready for Meta (Instagram), X, Telegram, and Google Play.

5. FRONTEND USER EXPERIENCE — 4 CORE SOC DASHBOARD SCREENS

- **Screen 1 — SOC Command Center:** Live KPI strip (Monitored Brands, Total Accounts Scanned, Critical Threats Blocked) + Animated Threat Radar.
- **Screen 2 — Real-Time Deep Scanner:** Input bar for Brand Name (e.g., *ByteXL*) and Suspect Handle (e.g., *@bytexl_careers_official*) with live sub-second telemetry.
- **Screen 3 — Split Forensic Comparison View:** Left side shows Verified Official Brand; Right side shows Imposter with highlighted red scam bio phrases, bounding-box logo match, and animated radial gauge (*96% CRITICAL*).
- **Screen 4 — Automated Enforcement Drawer:** Interactive legal takedown preview with 1-click PDF export and Abuse Desk email copy.

6. TECHNOLOGY STACK & DEPLOYMENT ARCHITECTURE

Layer	Technology Stack	Architectural Justification
Frontend UI	Next.js 15 / React 19 + Tailwind CSS v4 + shadcn/ui + Framer Motion	Responsive dark-mode Security Operations Center (SOC) aesthetic; deployed on Vercel Edge.
Backend API	Python 3.11+ / FastAPI (Async) + Pydantic v2	High-throughput REST endpoints (*/api/v1/scan*, */api/v1/takedown*) with strict schema validation.
AI & Vector Core	ONNX Runtime (*.onnx*) + Meta FAISS (*faiss-cpu*) + Levenshtein	Lightweight (<40MB) cross-platform neural inference without heavy GPU overhead; sub-30ms total scan time.
Training & Dev	Google Colab Pro (PyTorch *.pt* export) + Antigravity (AGY) IDE	Rapid model training/quantization in Colab paired with full-stack integration in AGY IDE.

7. EXECUTION ROADMAP & HACXLERATE 2026 EVALUATION ALIGNMENT

Phase / Role	Key Deliverables & Milestones	Evaluation Parameter Satisfied
Phase 1 (Hrs 0–6) Architecture & ML	Curate brand dataset, extract CLIP embeddings on Colab, export **.onnx** & FAISS **.index**.	**Technical Execution & Innovation:** Genuine vector similarity + neural pipeline.
Phase 2 (Hrs 6–16) Backend & SOC UI	Build FastAPI */scan* & */takedown* routes; build 4-screen Next.js dark-mode SOC dashboard.	**Functionality & UX:** Intuitive end-to-end scan flow passing objective checks.
Phase 3 (Hrs 16–24) Deploy & Verify	Deploy live frontend/backend URLs, verify GitHub documentation, test all acceptance criteria.	**Deployment & Scalability:** Live accessible production link + B2B SaaS readiness.

8. THE 3-MINUTE JUDGE DEMO & PITCH FLOW

- **0:00 – 0:45 (The Crisis Hook):** Show how scammers steal brand trust and student/customer money in minutes while manual takedowns take 72+ hours.
- **0:45 – 1:45 (Live Multi-Modal Scan):** Enter *ByteXL* and scan *@bytexl_support_official*. Show FAISS matching the spoofed logo (94%), NLP catching UPI fee extortion in the bio, and Levenshtein flagging handle typosquatting → **96% CRITICAL THREAT**.
- **1:45 – 2:30 (1-Click Takedown Knockout):** Click *'Generate Legal Takedown'* to produce a complete Cease & Desist PDF and abuse desk dispatch in 1 second.
- **2:30 – 3:00 (B2B Scale for Kampus.vc):** Highlight sub-second FAISS scaling across millions of profiles and enterprise SaaS market readiness.